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Unsupervised health indicator construction by a new Gaussian-student's t-distribution mixture model and its application

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ABSTRACT

The equipment's remaining useful life (RUL) must be accurately estimated to guarantee its reliable operation. As a crucial part of data-driven RUL prediction, the health indicator (HI) construction method employing the distribution discrepancies can represent the variation trend of health conditions. However, the existing Gaussian mixture model based HI construction method cannot accurately estimate the long-tail distribution characteristics in some degradation data. Moreover, it cannot comprehensively mine the distribution characteristics of degradation data by leveraging different types of distributions. A novel Gaussian-student's t-distribution mixture model (GSMM) that simultaneously considers Gaussian distribution and student's t-distribution is developed in this work to estimate the distributions of normal and degradation data. Next, the distribution contact ratio metric (DCRM) is applied to measure the discrepancies between the baseline distribution of normal data and the distributions of test data at different moments. The bearing HI can be constructed with the acquired DCRMs. Finally, the effectiveness and merit of the developed HI construction approach are validated by two bearing life-cycle datasets. The experimental results illustrate that the GSMM-based HI performs better than other classical and state-of-the-art HIs. Additionally, the constructed HI is more suitable for bearing RUL prediction.

1. Introduction

In modern industry, various rotating machineries have been generally applied for industrial production, such as machine tools, wind turbines, and aero-engines [1–3]. These mechanical equipment will inevitably degrade with service time and a harsh working environment, and some faults will occur [4]. As a critical mechanical component, the health monitoring of the bearing is crucial to the safe operation of the equipment [5]. To ensure the efficient production of equipment and reduce costly maintenance costs, it is essential to dynamically assess the performance degradation of the bearing and accurately estimate the remaining useful life (RUL) of the bearing [6]. Therefore, a new health indicator (HI) construction approach is proposed in this work by taking bearing as the research object. The generated HIs can be utilized for estimating the bearing RULs.

The existing methods for RUL prediction can be broadly categorized into physics-based and data-driven methods [7,8]. The physics-based methods can acquire accurate RULs of machines by establishing mathematical or physical models with known fault mechanisms. However,

constructing appropriate degradation models for complex systems by such methods can be difficult. The data-driven approaches have been explored for accurate RUL prediction to address the limitation of physics-based methods and realize the RUL estimation of complex systems. They can directly mine degradation information from monitoring data using shallow machine learning or deep learning techniques.

In recent years, various data-driven methods have been proposed for predicting the RUL of rotating machinery. Qin et al. [9] presented a variant of the capsule network for accurately predicting the RUL of machinery. This model can learn some significant dynamics information from the monitoring data. Zhang et al. [10] developed a two-stage neural network-based model to estimate the bearing RULs. Jin et al. [11] built a novel network for estimating the RULs of rotating machineries. Wei et al. [12] introduced a novel graph neural network to accurately estimate the RUL of bearing. Jiang et al. [13] proposed a novel convolutional dual-channel Transformer network to predict the RULs of rolling bearings. Zhang et al. [14] designed a trend-augmented and temporal-featured Transformer network for RUL prediction of rotating machineries. This improved Transformer network contains

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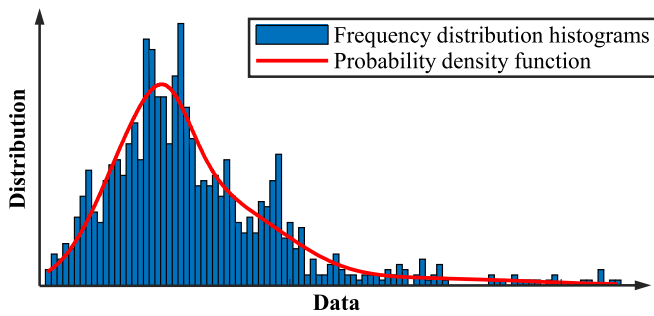


Fig. 1. The data distribution of degradation signals.

trend augmentation module and time-feature attention module. Shang et al. [15] introduced a novel data augmentation framework utilizing a dense convolutional regression network for RUL prediction. This framework was designed to make full use of the information present in the limited labeled data. In the existing data-driven RUL prediction methods, the prediction framework consists of data acquisition, HI construction (feature extraction), and RUL prediction [4]. In the first part, the monitoring data containing degradation information are collected via the installed sensors. The second part automatically extracts the HIs or degradation features from the acquired monitoring data. The RULs are accurately predicted based on the constructed HIs or the extracted features in the third part.

The HI construction (feature extraction) part is essential to RUL prediction, significantly affecting the prediction accuracy of data-driven method. In addition, the constructed HIs can assess the performance degradation of rotating machinery. Thus, constructing an appropriate HI is critical for the data-driven RUL prediction methods. Nowadays, many HI construction methods have been explored to assess the performance degradation of rotating machinery quantitatively. The constructed HIs can be classified into two distinct categories: physical HIs (PHIs) and virtual HIs (VHIs) [16]. In PHIs, the root mean square (RMS) is frequently utilized to describe the deterioration trend of rotating machinery. In addition, the RMS can also be directly leveraged as an HI for time-series-based RUL prediction. Klausen et al. [17] considered RMS as an HI for estimating bearing RULs. Moreover, Qin et al. utilized the RMS as a bearing HI [18], and the RULs were estimated using a gated dual attention unit network. Although RMS can monitor the machine's health condition through clear physical meaning, it is an ineffective HI for comprehensively representing the degradation trajectory of machine in some cases. This is mainly because RMS is just a time-domain statistical feature, and it is insensitive to initial weak fault. In addition, RMS might not effectively capture the localized or subtle changes in the frequency domain that could be indicative of early mechanical deterioration.

The VHIs are explored to monitor the degradation trajectory of the machine more comprehensively. The VHIs without specific physical meanings can be generated from several extracted PHIs via common dimension reduction algorithms or some optimization methods [19,20]. Zhao et al. [21] utilized principal component analysis (PCA) to construct a bearing's HI by fusing multiple time–frequency domain features. Chen et al. [22] applied kernel principal component analysis (KPCA) to generate HI and represent the degradation trajectory of machinery. An isometric mapping (ISOMAP) method was adopted by Qin et al. [23] to construct the gear's HI from some statistical features. These VHIs based on dimensionality reduction are constructed by performing dimensionality reduction on the selected signal features. The prior knowledge is required to extract some useful features as the input of the dimensionality reduction algorithms, such as PCA and KPCA.

Deep learning-based HI construction method was developed to directly generate HI from the collected life-cycle data without prior knowledge. The method can construct HIs in a supervised or unsupervised manner. In supervised HI construction methods, the HIs of machines can be constructed via some linear and non-linear HI labels. Guo

et al. [24] presented a supervised HI construction approach to generate bearing's HI via a linear HI label. Chen et al. [25] developed a health monitoring method based on convolution and recurrent neural networks. Cheng et al. [26] designed a nonlinear deterioration index as an HI label to construct a bearing's HI via a convolutional neural network. However, the constructed HIs may not adequately represent the practical degradation trajectory because the designed HI labels are inconsistent with the actual situation. On the contrary, unsupervised HI construction methods can automatically generate HIs without HI labels or prior knowledge. Thus, such methods are more suitable for HI construction in practical engineering applications than supervised HI construction methods.

Some investigations regarding unsupervised HI construction have been conducted in the HI construction field. Chen et al. [27] utilized a new auto-encoder model to construct bearing's unsupervised HI. Mao et al. [28] built a novel auto-encoder-based HI construction approach for the performance degradation assessment of a metro wheel. Li et al. [29] presented a new data fusion model for constructing HIs of aero-engines. Guo et al. [30] introduced an unsupervised HI construction approach using monitoring data under health condition for model training. Nevertheless, the existing unsupervised HI construction approaches can be employed to some application scenarios where the amount of monitoring data is sufficient. In addition, the quality of the training data is significant for these methods. However, it may be uneasy to collect adequate monitoring data in the industry, and the quality of the collected monitoring data might be poor.

As a popular probabilistic model, the Gaussian mixture model (GMM) has been employed to generate the HI from insufficient monitoring data. GMM can estimate the unknown data distribution with the expectation maximization (EM) algorithm. Some investigations about GMM-based HI construction have been conducted to construct machines' HIs from the data distribution perspective. Hong et al. [31] introduced a new bearing degradation performance evaluation approach by means of GMM and PCA. Rai and Upadhyay [32] utilized GMM and Jensen–Rényi divergence to generate the bearing's HI from vibration signals. He and Guo [33] explored an improved GMM to estimate the monitoring data distribution for HI construction. Zhou et al. [34] proposed a distribution contact ratio metric (DCRM) to measure the discrepancies between baseline distribution and distributions of life-cycle data at different moments. The GMM was applied to estimate the distribution of life-cycle data. Wen et al. [35] presented a new HI construction approach for early fault detection of bearings using GMM and Wasserstein distance. Notably, the GMM can only estimate data distribution of life-cycle data by combining various Gaussian distributions. Moreover, GMM may not accurately estimate the distributions of degradation data because the distributions of some data have long-tail characteristics [36–38] caused by the anomaly shown in Fig. 1. These distribution characteristics in Fig. 1 are not the same as the Gaussian distribution characteristics. Thus, exploring a novel mixture model for comprehensively representing the distribution characteristics of degradation data is essential.

Considering the student's t-distribution has a long-tail distribution property, a student's t-distribution mixture model (SMM) that combines various student's t-distributions is first applied to mine the long-tail distribution characteristics in the degradation data. However, the SMM may not be suitable for estimating the Gaussian features prevalent in the life-cycle data, particularly for the normal samples that do not have long-tail characteristics. Therefore, the Gaussian-student's t-distribution mixture model (GSMM) is further developed to comprehensively and accurately estimate the distributions of full life-cycle data. The GSMM can estimate distribution from the perspective of different types of distributions. Choosing the student's t-distribution to fit the long-tail distribution has several advantages over other distributions. 1) The student's t-distribution has a distinct tail compared to the Gaussian distribution, allowing for the more accurate capture of long-tail distribution characteristics. In addition, the student's t-distribution is more

robust than Gaussian distribution, providing greater tolerance to outliers or extreme values in the observation data. 2) Compared to other distributions, such as Cauchy distribution, Laplace distribution, the student's t-distribution is more flexible in adjusting the tail shapes by changing the degrees of freedom, which enables the proposed GSMM to fit different shapes of long-tail distributions more adaptively. Therefore, the student's t-distribution is more robust and flexible for fitting the long-tail distribution than Gaussian or other distributions [37,38]. The DCRM is an indicator for measuring distribution discrepancy. It has been proposed in Ref. [34], and its superiority for HI construction of bearing has also been validated in Ref. [34]. Therefore, the DCRM is applied to measure the discrepancies between the baseline distribution and the distributions of life-cycle data at different moments. The HI is constructed via the obtained distribution discrepancies. The GSMM-based HI can describe rotating machinery's degradation trajectory and health status. Moreover, the HI can improve the accuracy of the life estimation approach. The performance of the developed GSMM-based HI is verified via bearing datasets in the experimental section. The main contributions of this paper are summarized as follows:

- (1) A GSMM is first put forward by considering the Gaussian distribution characteristics and student's t-distribution characteristics simultaneously, and it can better estimate the distributions of both normal and degraded bearing signals than the existing GMM and SMM. The related parameter updating formulas of GSMM are also derived.
- (2) A novel unsupervised HI construction method is proposed to generate HI for quantitatively representing the degradation process via the developed GSMM and a distribution discrepancy metric named DCRM, which can measure the discrepancies between the baseline distribution and distributions of the life-cycle samples.
- (3) The proposed HI is successfully applied to the RUL prediction via two life-cycle datasets of bearings. The contrast experiment demonstrates that the presented GSMM-based HI is superior to other HIs, effectively improving the precision of RUL prediction.

The rest of this paper is organized as follows. Section 2 presents the fundamental theory of this work. Section 3 introduces the GSMM-based HI construction method. The effectiveness and superiority of the GSMM-based method are experimentally validated in Sections 4 and 5. Finally, Section 6 concludes this paper.

2. Fundamental theory

2.1. Gaussian mixture model

As a powerful probabilistic model, GMM can combine multiple Gaussian distributions to represent complex data distribution. The GMM has been widely utilized for clustering and probability density estimation tasks. The density function of GMM is defined as follows:

$$P(\mathbf{x}|\Theta) = \sum_{m=1}^M \pi_m \varphi(\mathbf{x}|\theta_m) \quad (1)$$

where $\mathbf{x}$ is p -dimensional observation data; $\Theta = (\pi_1, \pi_2, \dots, \pi_M; \theta_1, \theta_2, \dots, \theta_M)$ denotes the parameter set of GMM; M is the number of the Gaussian components in GMM; π_m represents the relative importance of the m -th component in the overall Gaussian mixture; $\varphi(\mathbf{x}|\theta_m)$ denotes Gaussian density function defined as follows:

$$\varphi(\mathbf{x}|\theta_m) = (2\pi)^{-\frac{p}{2}} |\Sigma_m|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_m)^T \Sigma_m^{-1} (\mathbf{x} - \mu_m)\right) \quad (2)$$

where μ_m denotes the mean vector; Σ_m denotes the covariance matrix. The parameters of GMM can be iteratively updated using EM algorithm.

2.2. Student's t-distribution mixture model

Although GMM can estimate some complex unknown distributions using a mixture of multiple Gaussian distributions, it cannot be utilized to estimate specific data distributions, especially the long-tail distributions. This is because the GMM assumes that the data distribution with Gaussian characteristics is unsuitable for parameter estimation of long-tail distributions. To address this issue, SMM is designed to estimate the distribution of data with long-tail characteristics. The density function of SMM is given as follows:

$$P(\mathbf{x}|\Theta) = \sum_{l=1}^L \pi_l \psi(\mathbf{x}|\theta_l) \quad (3)$$

where $\mathbf{x}$ denotes p -dimensional observed data, and $\Theta = (\pi_1, \pi_2, \dots, \pi_L; \theta_1, \theta_2, \dots, \theta_L)$ is the parameter set of SMM; L is the number of components of SMM; π_l represents the relative importance of the l -th component in SMM; $\psi(\mathbf{x}|\theta_l)$ is student's t-distribution defined as follows:

$$\psi(\mathbf{x}|\theta_l) = \frac{\Gamma\left(\frac{\nu_l+p}{2}\right) |\Omega_l|^{-\frac{1}{2}}}{(\pi\nu_l)^{\frac{p}{2}} \Gamma\left(\frac{\nu_l}{2}\right) \left[1 + \frac{\delta(\mathbf{x}|\mu_l, \Omega_l)}{\nu_l}\right]^{\frac{\nu_l+p}{2}}} \quad (4)$$

where

$$\delta(\mathbf{x}|\mu_l, \Omega_l) = (\mathbf{x} - \mu_l)^T \Omega_l^{-1} (\mathbf{x} - \mu_l) \quad (5)$$

in Eq. (5), δ represents the Mahalanobis squared distance between $\mathbf{x}$ and μ_l ; μ_l denotes the mean vector; Ω_l denotes the correlation matrix; ν_l denotes the degrees of freedom; Γ is the Gamma function. In order to obtain the optimal parameters of SMM, the EM method is employed to iteratively update the parameters in SMM.

3. HI construction method based on GSMM

The GSMM is first developed based on GMM and SMM. The GSMM can estimate data distributions of full life-cycle data from a mixture of various types of distributions. The bearing HI is constructed according to the estimated baseline distribution of normal life-cycle data and the estimated distributions of life-cycle data at different moments using DCRM to measure the distribution discrepancies between them.

3.1. Gaussian-student's t-distribution mixture model

The GMM can estimate complex distribution by combining finite Gaussian distributions. Thus, the GMM has been widely utilized to estimate the distribution of mechanical vibration signals for health monitoring. However, the GMM may be unsuitable for estimating the distribution of degradation data due to the long-tail characteristics caused by the anomaly. Therefore, the SMM can be employed to estimate the distributions of degradation signals. The SMM can automatically estimate the distributions of degradation data by combining multiple student's t-distributions. Nevertheless, the SMM cannot estimate the distributions of normal data that do not have long-tail characteristics. To address the aforementioned limitations, the GSMM is proposed for the HI construction of bearings by estimating normal and degradation data distributions simultaneously. The density function of GSMM is given as follows:

$$P(\mathbf{x}|\Theta) = \sum_{m=1}^M \pi_m \varphi(\mathbf{x}|\theta_m) + \sum_{l=1}^L \pi_l \psi(\mathbf{x}|\theta_l) \quad (6)$$

where $\mathbf{x}$ denotes the observation data; M and L are the number of Gaussian and student's t-distribution components, respectively. π is the positive relative importance that satisfies $\sum_{m=1}^M \pi_m + \sum_{l=1}^L \pi_l = 1$.

Given a dataset $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_B)$, the parameters of GSMM are

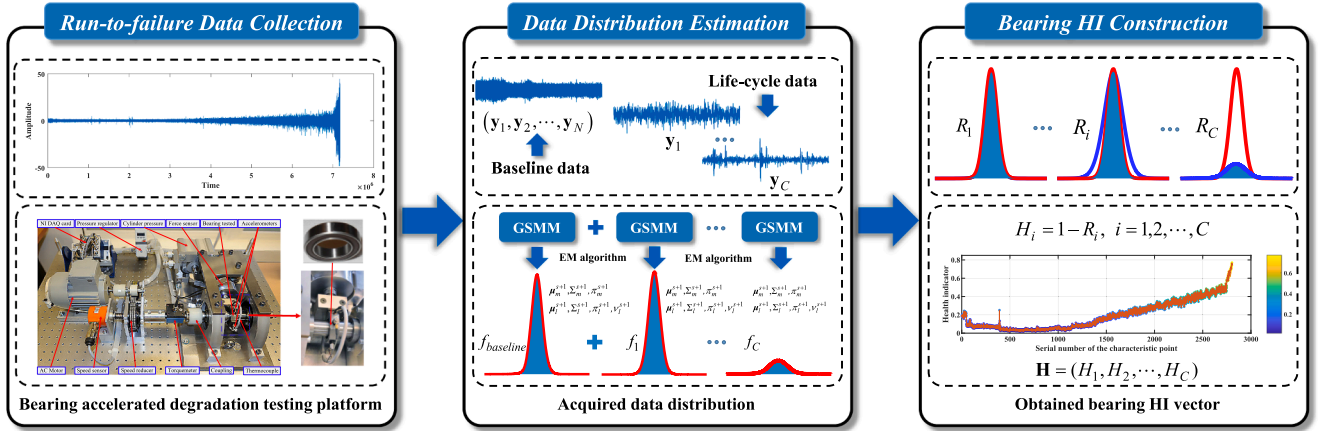


Fig. 2. The proposed HI construction framework.

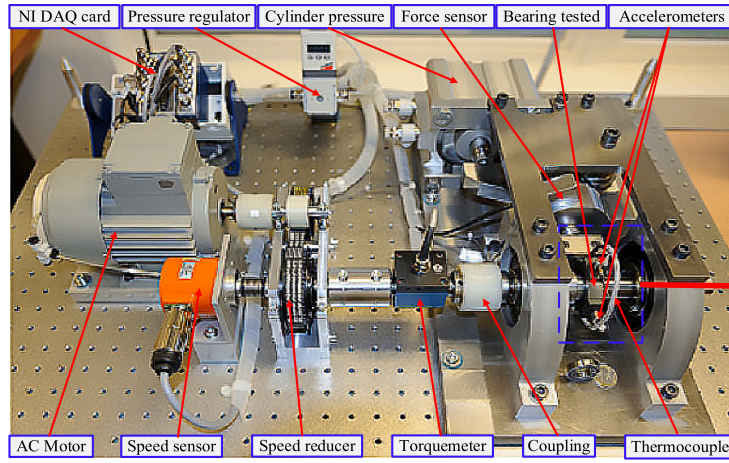


Fig. 3. PRONOSTIA platform.

Table 1

The detailed information of IEEE PHM 2012 bearing datasets.

Load (kN)	Rotating speed (rpm)	Datasets	Sample length
4	1800	Bearing1_1	2803
		Bearing1_3	2375
4.2	1650	Bearing2_1	911
		Bearing2_3	1955
5	1500	Bearing3_1	515
		Bearing3_3	434

estimated using EM algorithm. The EM method starts with an initial parameter estimation and iteratively updates the estimated value until convergence. In the expectation step (E-step) on the $(s+1)$ th iteration of the EM algorithm, the responsiveness of each component to each data is obtained via the estimated parameters in the s th iteration:

$$\gamma_{bm}^{s+1} = \pi_m^s \varphi(\mathbf{x}_b | \theta_m^s) / \left(\sum_{m=1}^M \pi_m^s \varphi(\mathbf{x}_b | \theta_m^s) + \sum_{l=1}^L \pi_l^s \psi(\mathbf{x}_b | \theta_l^s) \right) \quad (7)$$

$$\gamma_{bl}^{s+1} = \pi_l^s \psi(\mathbf{x}_b | \theta_l^s) / \left(\sum_{m=1}^M \pi_m^s \varphi(\mathbf{x}_b | \theta_m^s) + \sum_{l=1}^L \pi_l^s \psi(\mathbf{x}_b | \theta_l^s) \right) \quad (8)$$

The weight expectation for the student's t-distribution component in GSMM is calculated by the following formula:

$$\omega_{bl}^{s+1} = \frac{\nu_l^s + p}{\nu_l^s + (\mathbf{x}_b - \mu_l^s)^T (\Omega_l^s)^{-1} (\mathbf{x}_b - \mu_l^s)} \quad (9)$$

In the maximization step (M-step) of the $(s+1)$ th iteration, the parameters of GSMM are iteratively updated via the following formulas:

$$\mu_m^{s+1} = \sum_{b=1}^B \gamma_{bm}^{s+1} \mathbf{x}_b / \sum_{b=1}^B \gamma_{bm}^{s+1} \quad (10)$$

$$\Sigma_m^{s+1} = \sum_{b=1}^B \gamma_{bm}^{s+1} (\mathbf{x}_b - \mu_m^{s+1})(\mathbf{x}_b - \mu_m^{s+1})^T / \sum_{b=1}^B \gamma_{bm}^{s+1} \quad (11)$$

$$\pi_m^{s+1} = \sum_{b=1}^B \gamma_{bm}^{s+1} / B \quad (12)$$

$$\mu_l^{s+1} = \sum_{b=1}^B \gamma_{bl}^{s+1} \omega_{bl}^{s+1} \mathbf{x}_b / \sum_{b=1}^B \gamma_{bl}^{s+1} \omega_{bl}^{s+1} \quad (13)$$

$$\Omega_l^{s+1} = \sum_{b=1}^B \gamma_{bl}^{s+1} \omega_{bl}^{s+1} (\mathbf{x}_b - \mu_l^{s+1})(\mathbf{x}_b - \mu_l^{s+1})^T / \sum_{b=1}^B \gamma_{bl}^{s+1} \quad (14)$$

$$\pi_l^{s+1} = \sum_{b=1}^B \gamma_{bl}^{s+1} / B \quad (15)$$

The degrees of freedom ν_l^{s+1} for the student's t-distribution is

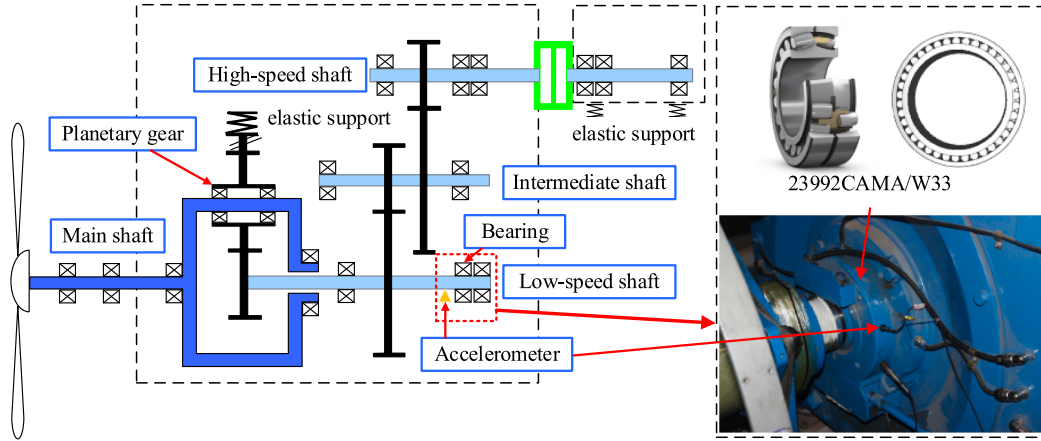


Fig. 4. The simplified structure diagram of wind turbine gearbox.

Table 2

The detailed information of WTGB datasets.

Dataset	J1	J2	J3	J4	J5	J6
Sample length	260	128	100	108	372	396

calculated as the solution to this equation:

$$\log\left(\frac{\nu_l^{s+1}}{2}\right) - \phi\left(\frac{\nu_l^{s+1}}{2}\right) + 1 - \log\left(\frac{\nu_l^s + p}{2}\right) + \frac{\sum_{b=1}^B \gamma_{bl}^{s+1} (\log \omega_{bl}^{s+1} - \omega_{bl}^{s+1})}{\sum_{b=1}^B \gamma_{bl}^{s+1}} + \phi\left(\frac{\nu_l^s + p}{2}\right) = 0 \quad (16)$$

where $\phi(x) = \frac{\partial(\ln \Gamma(x))}{\partial x}$ denotes the digamma function.

Compared with the Gaussian-student's t mixture distribution that only combines a Gaussian distribution and a student's t-distribution in Ref. [39], the Gaussian distribution and student's t-distribution are

utilized as the components of the proposed GSMM for mining the distribution characteristics of degeneration vibration signals. The number of Gaussian distributions and that of student's t-distributions are determined by Bayesian information criterion (BIC), and the parameters of GSMM are estimated by EM algorithm. However, the parameters of Gaussian-student's t mixture distribution are estimated using variational Bayesian method, and this type of mixture distribution is mainly applied to improve the estimation accuracy of Kalman filter.

3.2. Proposed HI construction method based GSMM

The framework of the developed HI construction methodology mainly comprises three parts: run-to-failure data collection, data distribution estimation, and bearing HI construction, as shown in Fig. 2. Firstly, the monitoring data are collected by the installed sensors. The acquired data is preprocessed when there is obvious noise in the raw data. Secondly, the GSMM is constructed for distribution estimation of the collected monitoring data. The baseline distribution of normal data and distributions of entire run-to-failure data are estimated via the

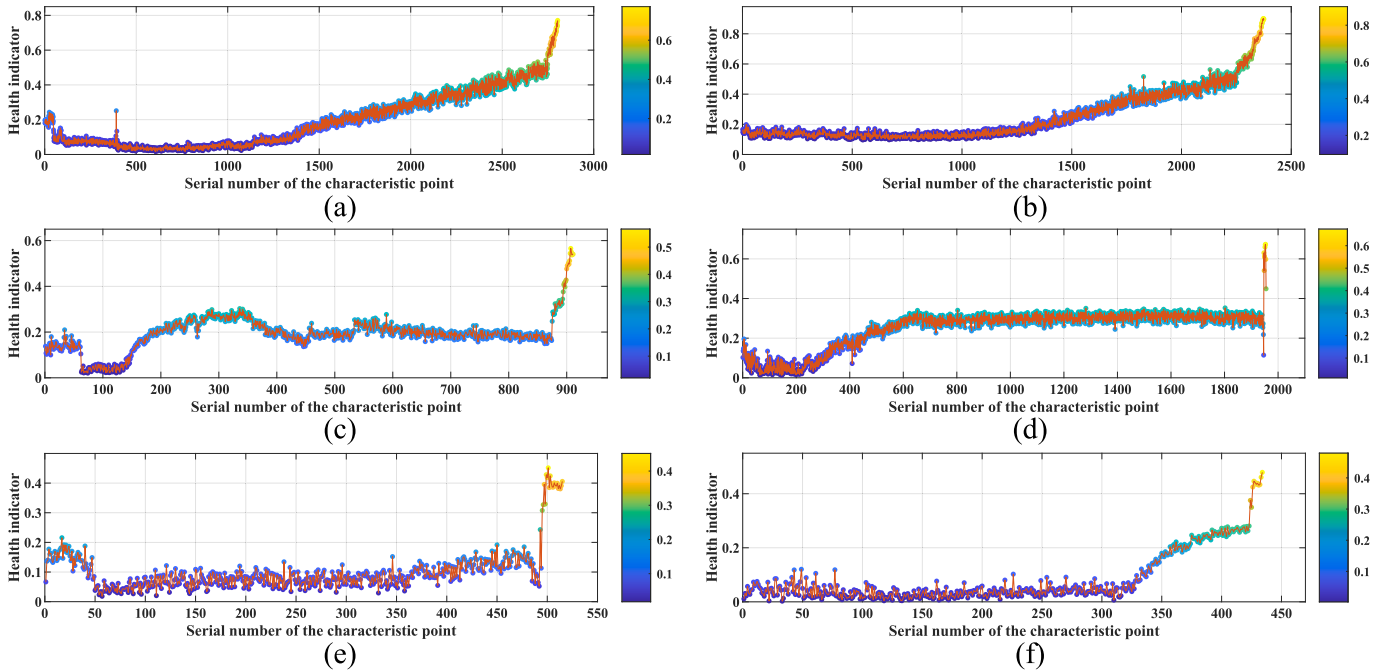


Fig. 5. The constructed GSMM-based HIs for IEEE PHM 2012 bearing datasets. (a) Bearing1_1, (b) Bearing1_3, (c) Bearing2_1, (d) Bearing2_3, (e) Bearing3_1, (f) Bearing3_3.

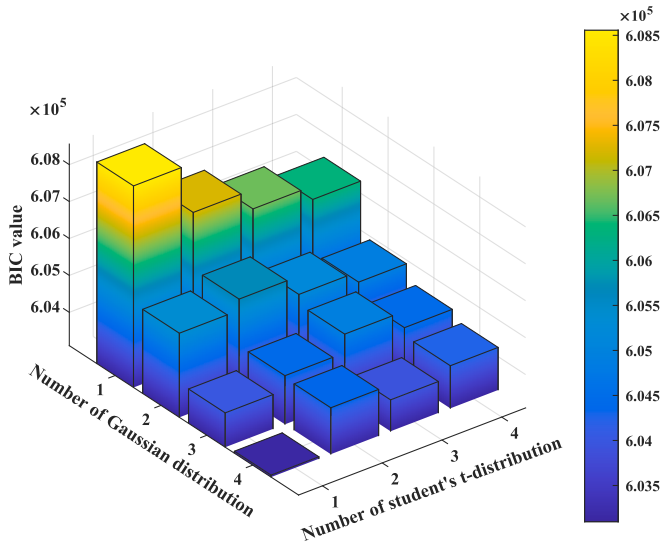


Fig. 6. The BIC values obtained by different number of components in GSMM.

constructed GSMM. Finally, the bearing's HI is generated by calculating the discrepancies between baseline distribution and all distributions of run-to-failure data.

The detailed descriptions of the developed HI construction framework are given as follows:

- (1) *Run-to-failure Data Collection*: The run-to-failure data (i.e., vibration signals) are acquired by the particular data acquisition system under preset working conditions. The obtained dataset is $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_C)^T$, $\mathbf{x}_i = (x_1, x_2, \dots, x_c)$, where C is the number of samples and c is the data length of a single sample.
- (2) *Data Distribution Estimation*: The GSMM is built by combining multiple Gaussian distributions and student's t-distributions. The BIC is a criterion used for model selection, particularly in the fields of statistics and machine learning. It aims to balance the goodness-of-fit and the complexity of models to select the most

suitable model. Therefore, the number of Gaussian distributions and student's t-distributions is set using the BIC, which is defined as follows:

$$BIC = k \ln(n) - 2 \ln(\hat{L}) \quad (17)$$

where k is the number of parameters to be estimated by model; n is the sample size; $\hat{L}$ is the maximum value of the model's likelihood function. As the number of Gaussian or student's t-distribution increases, the likelihood $\hat{L}$ of the model generally increases because the model becomes more flexible and can better capture the data characteristics. However, the increase in likelihood does not necessarily indicate the optimal model because it might lead to overfitting. In addition, as the number of Gaussian or student's t-distribution increases, the number of parameters k increases, leading to a larger penalty term in the calculation of BIC. This penalty discourages the selection of overly complex models. Therefore, the model with the lowest BIC value is considered to be optimal, as it offers the best balance between data goodness-of-fit and model complexity. Then, using the first N normal samples, the baseline distribution $f_{baseline}$ is estimated by the constructed GSMM method, while the data distribution $\mathbf{f} = (f_1, \dots, f_C)$ for all run-to-failure samples is estimated using the constructed GSMM.

- (3) *Bearing HI construction*: The discrepancies between baseline distribution $f_{baseline}$ and distributions of all monitoring samples $\mathbf{f} = (f_1, \dots, f_C)$ can be calculated by DCRM, which is defined as follows:

$$DCRM(f_{baseline} || f_i) = 1 - R_i, \quad i = 1, 2, \dots, C \quad (18)$$

where f_i is the distribution of i th monitoring sample; R_i denotes the overlapping area between $f_{baseline}$ and f_i . It can be calculated by the following formula:

$$R_i = \int \min[f_{baseline}(x), f_i(x)] dx \quad (19)$$

Finally, the bearing HI vector is obtained via calculating DCRMs to assess the bearing's performance degradation.

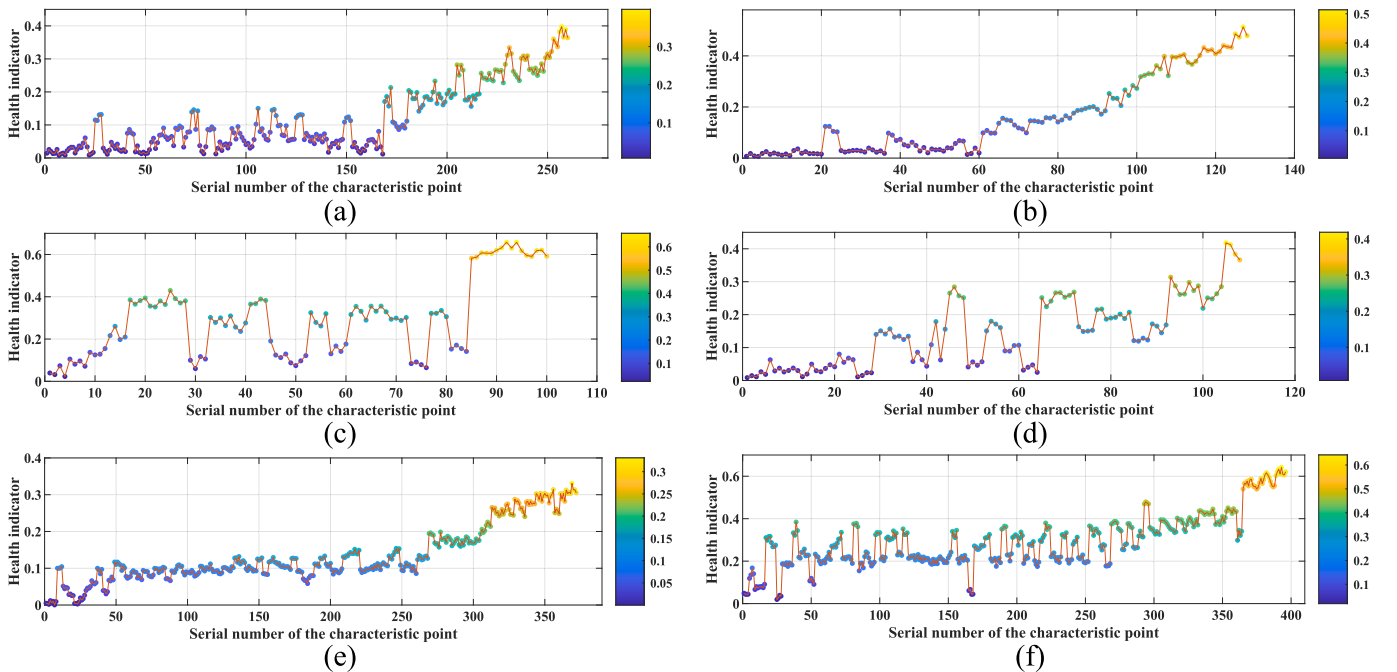


Fig. 7. The constructed GSMM-based HIs for WTGB datasets. (a) J1, (b) J2, (c) J3, (d) J4, (e) J5, (f) J6.

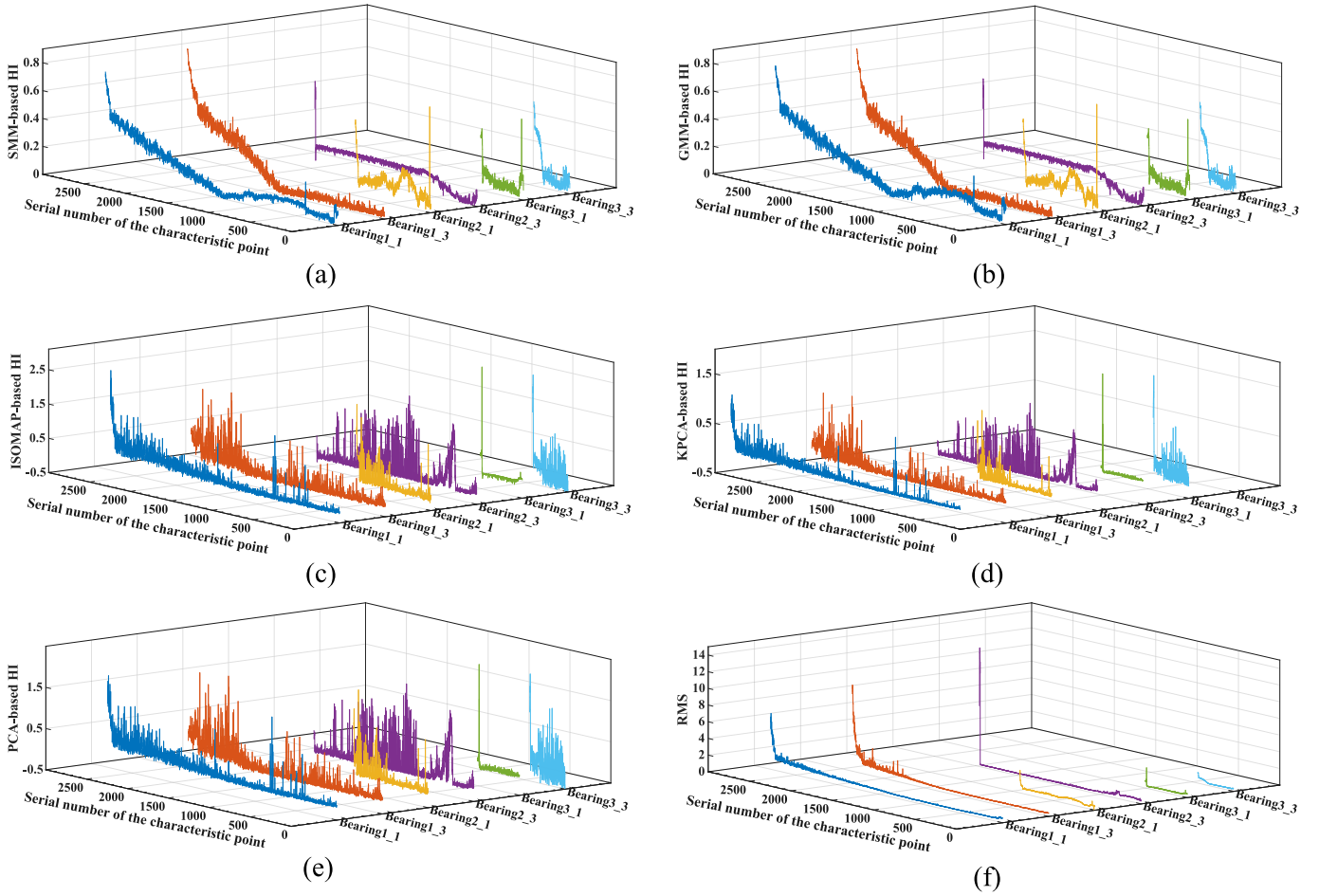


Fig. 8. HIs constructed by the comparative methods for IEEE PHM 2012 bearing dataset. (a) SMM, (b) GMM, (c) ISOMAP, (d) KPCA, (e) PCA, (f) RMS..

4. Experimental analysis and comparison

The experimental and actual bearing datasets are first introduced for HI construction. Then, the GSMM-based HIs are constructed to capture the deterioration trajectories of bearings via the developed HI construction approach. Finally, the effectiveness and merit of the GSMM-based HI are verified by comparing fusion and advanced HIs.

4.1. Bearing run-to-failure datasets description

- 1) *IEEE PHM 2012 bearing datasets*: the IEEE PHM 2012 bearing datasets [40] were provided by the FEMTO-ST Institute. The accelerated degradation experiments of bearings were carried out on the PRO-NOSTIA platform, as shown in Fig. 3. This platform mainly comprises rotating, loading, and measurement sections. An asynchronous motor and two shafts are located in the rotating section. A radial force was applied to tested bearings via a force actuator in the loading section. Lastly, vertical and horizontal data were acquired via vibration sensors in the measurement section. A total of 2560 samples were sampled every 10 s at a frequency of 25.6 kHz. The bearing test was stopped when the amplitude of the collected signals exceeded 20 g. In this work, six run-to-failure datasets are utilized to verify the efficacy of the developed HI construction method. The detailed information on these datasets is listed in Table 1.
- 2) *Wind turbine gearbox bearing (WTGB) datasets*: the WTGB datasets were acquired from the actual wind farm. The power of the wind turbine gearboxes was 2 MW, and the simplified structure diagram of the gearbox is displayed in Fig. 4. There is a single planetary gearbox transmission chain and a two-stage parallel gear transmission chain.

The monitoring data of the bearing for the low-speed shaft were collected by sensors. The signals were sampled at a sampling frequency of 25.6 kHz and a sampling time of 1.28 s. Each sample has the same sampling interval; four samples were obtained daily. The acquired signals were processed by wavelet denoising to reduce noise in raw signals. In this work, six monitoring datasets of WTGB are utilized to verify the developed HI construction method. The detailed information on these datasets is shown in Table 2.

4.2. Gsmm-based HI construction

The IEEE PHM 2012 bearing datasets are first applied to generate HIs of bearings. As shown in Fig. 5, the HIs of six bearings are constructed to verify the GSMM-based HI construction method. The first 200 normal samples are used to estimate the baseline distribution. The number of components of GSMM is selected via BIC to estimate the data distribution of vibration signals, with the maximum number of parameter estimation iterations being 100. The *K-means* method is employed for parameter initialization of GSMM. Taking Bearing1_1 as an example, the BIC values of GSMMs with different number of components are displayed in Fig. 6. It can be seen from this figure that the minimum BIC value can be achieved when the number of student's t-distributions is 1 and the number of Gaussian distributions is 4. As a result, the two hyper-parameters are respectively taken as 1 and 4.

According to Fig. 5, the constructed GSMM-based HIs can represent bearings' health status and degradation trajectory. In addition, the constructed HI values can be limited to [0, 1]. In the health stage, the HI values are close to zero, illustrating that the distribution of the current vibration signals is similar to the baseline distribution. In the

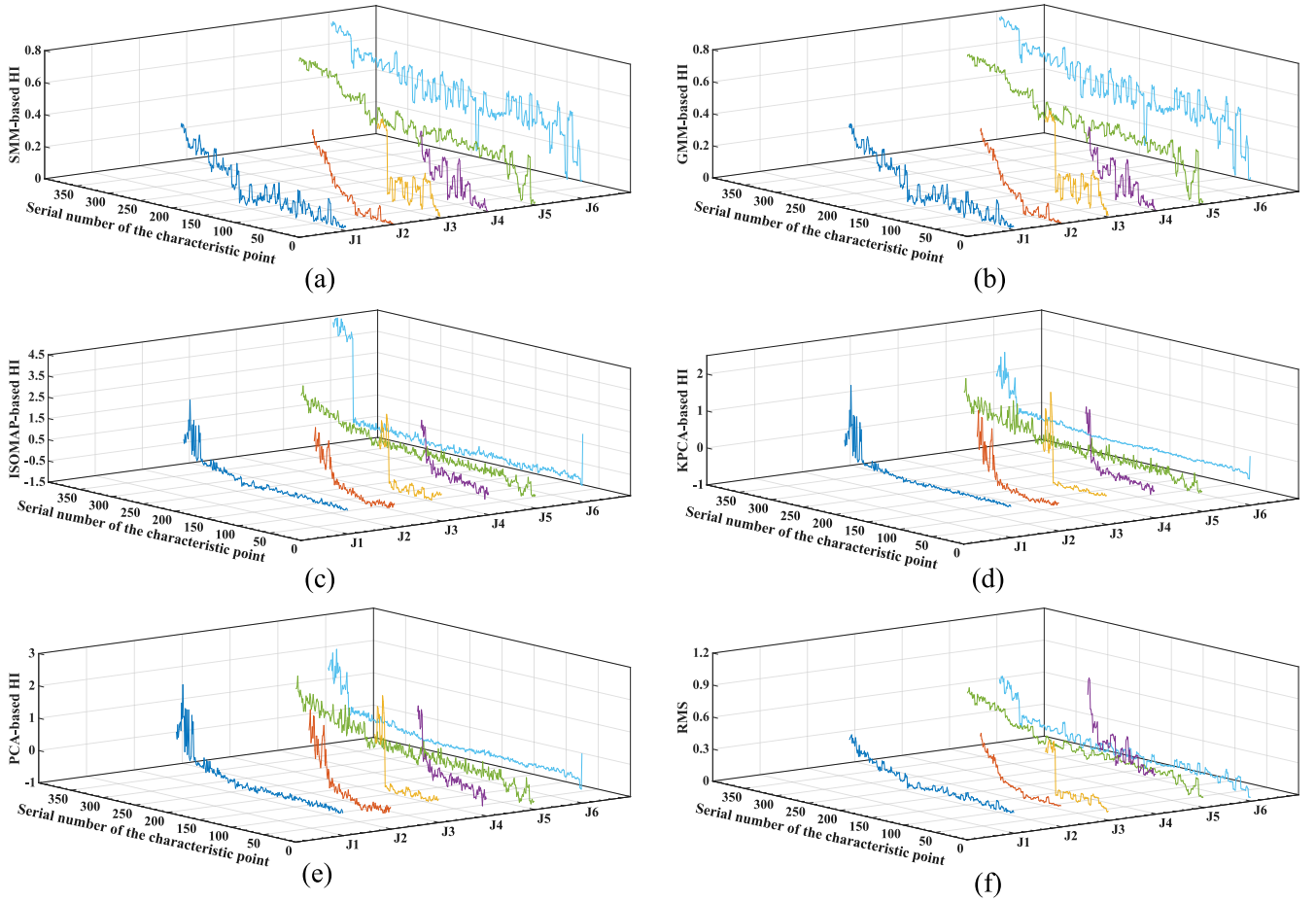


Fig. 9. HIs constructed by the comparative methods for WTGB datasets. (a) SMM, (b) GMM, (c) ISOMAP, (d) KPCA, (e) PCA, (f) RMS.

Table 3

The performance evaluation of bearing HIs for IEEE PHM 2012 bearing datasets.

Method	Bearing1_1					Bearing1_3					Bearing2_1				
	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR
GSMM	0.226	0.886	0.966	0.580	/	0.302	0.853	0.966	0.611	/	0.051	0.442	0.950	0.399	/
SMM	0.181	0.802	0.962	0.540	↑7.4 %	0.248	0.856	0.960	0.583	↑4.8 %	0.079	0.204	0.924	0.357	↑11.8 %
GMM	0.187	0.795	0.956	0.539	↑7.6 %	0.252	0.854	0.960	0.585	↑4.4 %	0.083	0.209	0.926	0.361	↑10.5 %
ISOMAP	0.111	0.686	0.927	0.471	↑23.1 %	0.085	0.561	0.777	0.388	↑57.5 %	0.224	0.478	0.846	0.461	↑13.4 %
KPCA	0.106	0.645	0.926	0.460	↑26.1 %	0.074	0.521	0.740	0.363	↑68.3 %	0.233	0.398	0.795	0.435	↑8.3 %
PCA	0.104	0.675	0.918	0.462	↑25.5 %	0.078	0.541	0.741	0.369	↑65.6 %	0.228	0.415	0.769	0.428	↑6.8 %
RMS	0.199	0.679	0.983	0.530	↑9.4 %	0.272	0.550	0.986	0.542	↑12.7 %	0.107	0.395	0.959	0.420	↑5.0 %
Method	Bearing2_3					Bearing3-1					Bearing3_3				
	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR
GSMM	0.159	0.779	0.930	0.514	/	0.195	0.376	0.888	0.439	/	0.330	0.704	0.914	0.580	/
SMM	0.153	0.725	0.947	0.506	↑1.6 %	0.175	0.358	0.887	0.425	↑3.3 %	0.265	0.735	0.905	0.551	↑5.3 %
GMM	0.142	0.733	0.947	0.502	↑2.4 %	0.175	0.366	0.887	0.427	↑2.8 %	0.265	0.733	0.904	0.550	↑5.5 %
ISOMAP	0.087	0.120	0.762	0.296	↑73.7 %	0.108	0.154	0.913	0.359	↑22.3 %	0.302	0.447	0.800	0.480	↑20.8 %
KPCA	0.090	0.113	0.773	0.299	↑71.9 %	0.038	0.151	0.820	0.295	↑48.8 %	0.237	0.338	0.797	0.425	↑36.5 %
PCA	0.088	0.116	0.766	0.297	↑73.1 %	0.054	0.175	0.812	0.306	↑43.5 %	0.233	0.318	0.736	0.401	↑44.6 %
RMS	0.085	0.021	0.980	0.341	↑50.7 %	0.229	0.065	0.917	0.403	↑8.9 %	0.030	0.640	0.946	0.427	↑35.8 %

degradation stage, the HI values are higher than zero, indicating that the distribution of the collected vibration signals differs from the baseline distribution. Consequently, some faults will lead to an abnormal bearing state. The HI value reaches the maximum in the failure stage, demonstrating that the bearing has been damaged and can no longer work.

The WTGB datasets are also applied to verify the performance of the GSMM-based approach. The first 5 normal samples are used to estimate the baseline distribution. The constructed HIs are shown in Fig. 7, revealing that the bearing degradation can be evaluated by the

generated HIs. Finally, Figs. 5 and 7 depict that the GSMM-based HI construction is effective in constructing bearing HI and assessing the performance degradation of a bearing. It can be seen from Fig. 7 that some HIs have local trend fluctuations. The main reason of these fluctuations is that the change of the rotating speed and load in WT affects the variation trend of HI. Additionally, the vibration amplitude also fluctuates under the almost same working conditions owing to some random factors of the system itself, such as random shape errors of machined parts, randomness of material properties, et al.

Table 4

The performance evaluation of bearing HIs for WTGB datasets.

Method	J1					J2					J3				
	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR
GSMM	0.699	0.825	0.844	0.768	/	0.748	0.913	0.916	0.831	/	0.556	0.569	0.536	0.553	/
SMM	0.691	0.817	0.827	0.757	↑1.5 %	0.685	0.905	0.918	0.799	↑4.0 %	0.414	0.615	0.637	0.521	↑6.1 %
GMM	0.683	0.823	0.831	0.755	↑1.7 %	0.701	0.909	0.918	0.808	↑2.8 %	0.414	0.617	0.636	0.521	↑6.1 %
ISOMAP	0.645	0.690	0.881	0.725	↑5.9 %	0.449	0.808	0.873	0.648	↑28.2 %	0.293	0.601	0.746	0.491	↑12.6 %
KPCA	0.529	0.587	0.878	0.645	↑19.1 %	0.370	0.733	0.860	0.590	↑40.9 %	0.313	0.579	0.724	0.489	↑13.1 %
PCA	0.606	0.658	0.885	0.700	↑9.7 %	0.449	0.798	0.870	0.645	↑28.8 %	0.313	0.608	0.739	0.500	↑10.6 %
RMS	0.690	0.805	0.889	0.773	↓0.6 %	0.638	0.861	0.928	0.770	↑7.9 %	0.353	0.637	0.716	0.519	↑6.6 %
Method	J4					J5					J6				
	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR	Mon	Cor	Rob	CI	PR
GSMM	0.981	0.783	0.717	0.862	/	0.639	0.868	0.899	0.763	/	0.544	0.754	0.836	0.674	/
SMM	0.990	0.773	0.702	0.860	↑0.2 %	0.623	0.871	0.889	0.752	↑1.5 %	0.524	0.705	0.814	0.647	↑4.2 %
GMM	0.981	0.787	0.721	0.864	↓0.2 %	0.601	0.867	0.890	0.741	↑3.0 %	0.519	0.718	0.820	0.649	↑3.9 %
ISOMAP	0.888	0.712	0.780	0.820	↑5.1 %	0.623	0.860	0.887	0.750	↑1.7 %	0.083	0.496	0.933	0.421	↑60.1 %
KPCA	0.850	0.636	0.776	0.785	↑9.8 %	0.601	0.821	0.845	0.718	↑6.3 %	0.418	0.608	0.940	0.613	↑9.9 %
PCA	0.869	0.701	0.782	0.809	↑6.6 %	0.623	0.854	0.867	0.742	↑2.8 %	0.423	0.654	0.937	0.623	↑8.2 %
RMS	0.944	0.762	0.753	0.850	↑1.4 %	0.628	0.863	0.902	0.757	↑0.8 %	0.494	0.704	0.892	0.655	↑2.9 %

Table 5

The average evaluation results for the two bearing datasets.

Method	Mon	Cor	Rob	CI	PR
GSMM	0.453	0.729	0.864	0.631	/
SMM	0.419	0.697	0.864	0.608	↑3.8 %
GMM	0.417	0.701	0.866	0.609	↑3.6 %
ISOMAP	0.325	0.551	0.844	0.526	↑20.0 %
KPCA	0.322	0.511	0.823	0.510	↑23.7 %
PCA	0.339	0.543	0.818	0.523	↑20.7 %
RMS	0.389	0.582	0.904	0.582	↑8.4 %

4.3. Comparative results of HI construction

The RMS, three fusion HIs constructed by PCA, KPCA, and ISOMAP, and two HIs generated via GMM and SMM are compared to demonstrate the superiority of the GSMM. For fusion HIs, 10 statistical features from time and frequency domains are fused to construct a virtual bearing HI. The constructed HIs by these comparison methods for IEEE PHM 2012 bearing and WTGB datasets are displayed in Figs. 8 and 9, respectively. From Figs. 5 and 7-9, it can be seen that some HIs constructed by the comparative methods have greater fluctuations compared with the proposed GSMM-based HI. Due to these fluctuations, the degradation process of bearing is not well characterized by these HIs, leading to the poor RUL prediction results.

Three evaluation indexes, including monotonicity (*Mon*), correlation (*Cor*), and robustness (*Rob*) [41], are used to assess the performance of the bearing's HI quantitatively. In addition, a composite index (*CI*) combining the above three indexes is employed to assess the HI performance comprehensively. The definition of *CI* is shown as follows:

$$CI = \theta_1 * Mon + \theta_2 * Cor + \theta_3 * Rob \quad (20)$$

where θ_1 , θ_2 , and θ_3 are set as 0.5, 0.2, and 0.3, respectively. In the published works, the weight parameters of *Mon*, *Cor*, and *Rob* in *CI* are generally set by experience alone. Currently, there is no a theory or rule to guide the set of these weight parameters. We think that *Mon*, *Cor*, and *Rob* play different roles in evaluating the performance of HI on RUL prediction. It follows that they are not considered to be equally important in this work. In general, the *Mon* of HI is more significant for the RUL prediction compared to *Cor* and *Rob*, therefore, the *Mon* is assigned a maximum weight, while the *Cor* and *Rob* are assigned relatively small weights [41].

The specific results of *Mon*, *Cor*, *Rob*, and *CI* for IEEE PHM 2012 bearing and WTGB datasets are presented in Tables 3 and 4, separately. To highlight the advantage of the proposed GSMM, the promoting rate (PR) of proposed HI construction method compared to other method is

calculated and given in Tables 3 and 4. The PR is defined as:

$$PR = \left(\frac{CI_{Proposed \ method} - CI_{Comparative \ method}}{CI_{Comparative \ method}} \right) \times 100\% \quad (21)$$

From Tables 3 and 4, we can note that the proposed method achieves the largest *CI*s in all the datasets except the Bearing2_1. For Bearing1_1, the PR results show that the *CI* of GSMM-based HI is 7.4 %, 7.6 %, 23.1 %, 26.1 %, 25.5 %, and 9.4 % higher than those of SMM-based HI, GMM-based HI, ISOMAP-based HI, KPCA-based HI and PCA-based HI and RMS. For Bearing2_1, GSMM is inferior to ISOMAP, KPCA, PCA and RMS, but outperforms SMM and GMM. It can be also seen from Tables III and IV that GSMM-based HI has a better monotonic property which is beneficial to time-series-based RUL prediction. Taking Bearing1_1 as an example, the *Mon* of GSMM-based HI is 0.226, which is larger than those of PCA-based HI, KPCA-based HI, and ISOMAP-based HI. The similar conclusion can be obtained from the *Mon* values for Bearing1_3, Bearing2_3, Bearing3_1, Bearing3_3, J1, J2, J3, J4, J5, and J6. Additionally, the *Mon* value of GSMM-based HI is larger than those of GMM-based HI and SMM-based HI for all bearing datasets except for J4, indicating that the proposed GSMM can more accurately estimate the distributions of vibration signals than GMM and SMM. This is mainly because the GMM and SMM only estimate data distribution from the perspective of single distribution. The *Cor* and *Rob* of GSMM-based HI are superior to or comparable to those of other HIs, that is, the proposed GSMM-based HI has good temporal correlation and robust performance, improving the prediction accuracy of the time-series-based approach.

To validate the robustness of the proposed GSMM-based HI, the average results of *Mon*, *Cor*, *Rob*, and *CI* for the IEEE PHM 2012 bearing and WTGB datasets are shown in Table 5. Obviously, the *Mon* and *Cor* of the GSMM-based HI are larger than those of other HIs. The *Rob* of GSMM-based HI is larger than those of the fusion HIs, while it is smaller than those of SMM-based HI, GMM-based HI, and RMS. Among these seven HIs, GSMM-based HI has the largest *CI*, and it is 3.8 %, 3.6 %, 20.0 %, 23.7 %, and 20.7 % higher than those of SMM-based HI, GMM-based HI, ISOMAP-based HI, KPCA-based HI and PCA-based HI and RMS. It implies that the proposed GSMM-based HI achieves a significant performance improvement compared to other HIs, especially the fusion HIs. Therefore, the proposed GSMM-based HI can better characterize the bearing degradation process and be more suitable for RUL prediction of bearings.

5. Application analysis and RUL prediction

The multi-hierarchical long-term memory augmented network (MLMA-Net) is first introduced for bearing's RUL prediction. Then, the prediction results are acquired by the constructed GSMM-based HIs.

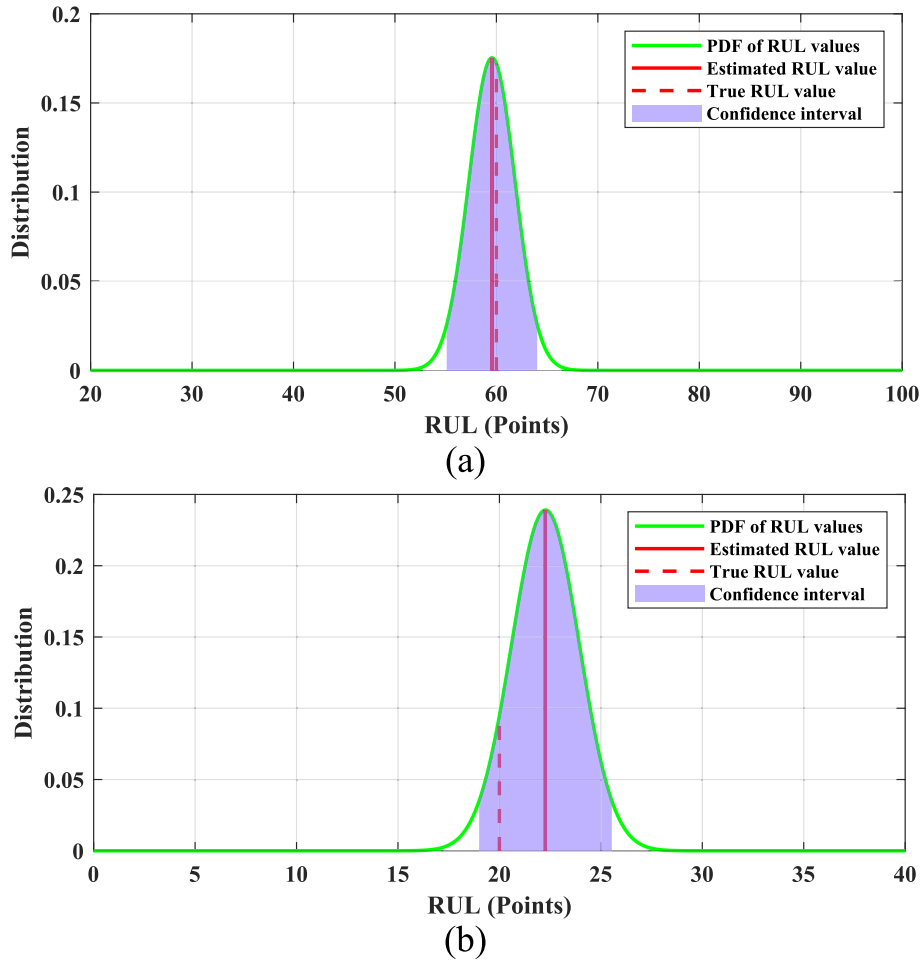


Fig. 10. The PDFs of RUL prediction. (a) bearing1_1, (b) J1.

Finally, the comparative results illustrate that the GSMM-based HI is more suitable for RUL prediction.

5.1. Mlma-net-based RUL prediction method

A data-driven RUL prediction method should be used to predict the RUL of bearing and verify the application of the GSMM-based HI. The RUL prediction method can extract time-varying deterioration information from the known HI values and estimate the unknown HI values. The long short-term memory network and its variants have been explored for bearing RUL prediction. In particular, the MLMA-Net has been successfully applied to RUL prediction through the specific gating mechanism. Thus, choosing the MLMA-Net as the prediction model in this work is appropriate. The MLMA-Net can accurately estimate the RULs of bearings and quantify the uncertainty of the MLMA-Net by the dropout technology. Some reasonable maintenance decisions and control schemes can be made based on the predicted RULs to improve production efficiency and reduce maintenance costs. More detailed information about MLMA-Net can be found in [42]. The final output of MLMA-Net is obtained using the cell state $\mathbf{h}_t$ as follows:

$$\begin{aligned}
 \mathbf{y}_t^1 &= \text{Dropout}(\mathbf{h}_t) \\
 \mathbf{y}_t^2 &= \text{Dropout}(\text{Relu}(\mathbf{W}_1 \mathbf{y}_t^1 + \mathbf{b}_1)) \\
 \mathbf{y}_t^3 &= \mathbf{W}_2 \mathbf{y}_t^2 + \mathbf{b}_2 \\
 \mathbf{y}_t &= \mathbf{W}_3 (\text{Dropout}(\mathbf{y}_t^3 + \mathbf{h}_t)) + \mathbf{b}_3
 \end{aligned} \tag{22}$$

where $\mathbf{W}$ and $\mathbf{b}$ are learnable weight and bias parameters, and $\mathbf{y}_t$ is the output value.

5.2. RUL prediction application of GSMM-based HI

The constructed HIs of bearing1_1 and J1 are employed to estimate the bearing RULs using the MLMA-Net-based method. A total of 1609 HI values from the 1140th value to the 2748th value are adopted for bearing1_1, and the last 60 HI values are predicted via the trained MLMA-Net. The first 240 HI values are leveraged to build a training dataset for J1, while the remaining 20 HI values are estimated via the MLMA-Net-based method.

The MLMA-Net is trained using the constructed bearing HIs, obtaining the RULs of bearing1_1 and J1. A total of 100 RULs are estimated by repeatedly testing the trained MLMA-Net based on bearing HI to guarantee the fairness and robustness of the estimated RULs. Finally, the mean value of the 100 RULs is used as the predicted bearing RUL. In the testing stage, the dropout operation is opened to quantify the uncertainty in MLMA-Net. The estimated RULs, actual RULs, the probability density functions (PDFs) of RUL, and the 95 % confidence interval are shown in Fig. 10.

According to Fig. 10, the MLMA-Net-based method can predict accurate bearing RULs based on the GSMM-based HIs and the estimated RULs with high confidence. In addition, the PDFs of the estimated RULs are beneficial for the preventive maintenance of bearings. The machinery downtime will be reduced, and the economic benefits will be promoted using the estimated PDFs. In conclusion, the GSMM-based HI is deemed suitable for RUL prediction. Additionally, the prediction accuracy of MLMA-Net-based method can be improved by the GSMM-based HI.

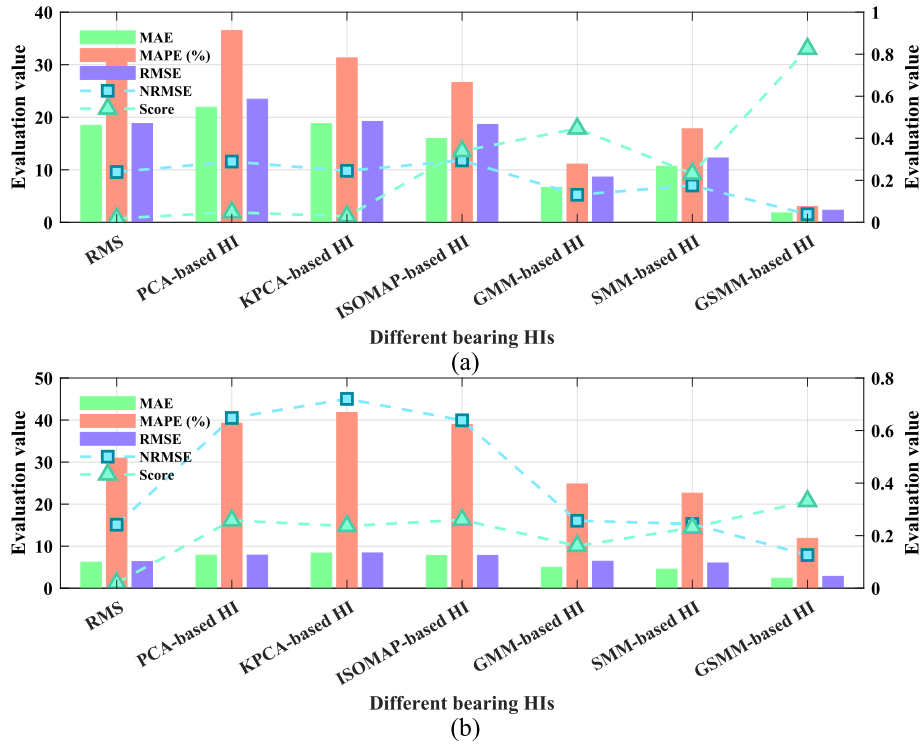


Fig. 11. Evaluation results. (a) bearing1_1, (b) J1.

5.3. Comparative results of RUL prediction

The RMS, fusion HIs, GMM-based HI, and SMM-based HI are employed for RUL prediction to demonstrate the superiority of the proposed GSMM-based HI. The prediction performance of HIs is evaluated by several evaluation metrics containing mean absolute error (MAE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), root mean square error (RMSE), and score. The evaluation results are displayed in Fig. 11.

According to Fig. 11, the GSMM-based HI has the highest score and lowest prediction error. A comparison of the evaluation results of GSMM-based HI, PCA-based HI, KPCA-based HI, and ISOMAP-based HI reveals that the presented GSMM-based HI has better prediction performance than traditional fusion HIs. In addition, the GSMM-based HI is superior to GMM-based and SMM-based HIs regarding five metrics, illustrating that the GSMM can generate better bearing HI for RUL prediction than GMM and SMM. Better results are mainly attributed to the GSMM's ability to estimate vibration signal distribution by combining different types of distributions. Finally, it can be concluded that the GSMM-based HI is better than other HIs and more appropriate for time-series-based RUL prediction.

6. Conclusions

A novel HI construction method was proposed in this work based on the GSMM, which is a variant of GMM that mines distribution characteristics of full life-cycle data from different types of distribution perspectives. This property is effective for estimating distributions of the bearing's vibration signals. The baseline distribution of normal data and distributions of life-cycle data at different moments were automatically estimated via the constructed GSMM. The distribution discrepancies between baseline distribution and distributions of entire life-cycle data were measured by DCRM to monitor the performance degradation of bearing quantitatively. The bearing HI was generated using the acquired distribution discrepancies. The effectiveness and merit of the proposed GSMM-based HI construction method were verified by two bearing

datasets. In addition, the constructed GSMM-based HI was applied for life prediction. The experimental results demonstrated that the GSMM-based HI is capable of representing the bearing's health condition and is superior to other fusion and advanced HIs. Moreover, the GSMM-based HI is more appropriate for improving RUL prediction accuracy than other HIs.

In the future work, it is worth exploring an unsupervised HI construction method based on GSMM under time-varying conditions. It is necessary to analyze the influence of time-varying operating conditions on the distribution parameters in GSMM. In addition, HIs for both incipient fault detection and prognosis under time-varying working conditions need to be researched.

CRedit authorship contribution statement

Dingliang Chen: Writing – original draft, Methodology, Investigation. **Yi Chai:** Visualization, Data curation. **Yongfang Mao:** Validation, Investigation. **Yi Qin:** Visualization, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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